

By letting $t \rightarrow \infty$, Fatou's lemma implies

$$\bar{L}(n)\mathbb{P}^x(\tau_{B_{r_0}(x_0)} > \tau_n) \leq \bar{L}(|x - x_0|), \quad x \in B_{r_0}^c(x_0).$$

Hence, by letting $n \rightarrow \infty$, we conclude that

$$\mathbb{P}^x(\tau_{B_{r_0}(x_0)} = \infty) \leq \frac{\bar{L}(|x - x_0|)}{\bar{L}(\infty)} < 1, \quad x \in B_{r_0}^c(x_0),$$

i.e., $\mathbb{P}^x(\tau_{B_{r_0}(x_0)} < \infty) > 0$ for all $x \in \mathbb{R}^d$.

We next show that $\{X(t)\}_{t \geq 0}$ is open-set irreducible. According to [21, Theorems 7.3.6 and 7.3.7], there exists a strictly positive function $q(t, x, y)$ on $(0, \infty) \times \bar{B}_{r_0}(x_0) \times \bar{B}_{r_0}(x_0)$, jointly continuous in t, x and y , and twice continuously differentiable in $x \in B_{r_0}(x_0)$, satisfying

$$\mathbb{E}^x[f(X(t))\mathbb{1}_{\{\tau_{\bar{B}_{r_0}^c(x_0)} > t\}}] = \int_{B_{r_0}(x_0)} q(t, x, y)f(y)dy,$$

for any $t > 0$, $x \in B_{r_0}(x_0)$ and any continuous, bounded function $f : \mathbb{R}^d \rightarrow \mathbb{R}$. By employing the dominated convergence theorem, the above relation holds also for $\mathbb{1}_O$, where $O \subseteq B_{r_0}(x_0)$ is an open set. Denote by $\mathcal{D}$ the class of all $B \in \mathcal{B}(B_{r_0}(x_0))$ (the Borel σ -algebra on $B_{r_0}(x_0)$) such that

$$\mathbb{P}^x(X_t \in B, \tau_{\bar{B}_{r_0}^c(x_0)} > t) = \int_B q(t, x, y)dy, \quad t > 0, \quad x \in B_{r_0}(x_0).$$

Clearly, $\mathcal{D}$ contains the π -system of open rectangles in $B_{r_0}(x_0)$, and forms a λ -system. By Dynkin's π - λ theorem, we conclude that $\mathcal{D} = \mathcal{B}(B_{r_0}(x_0))$. Consequently, for any $t > 0$, $x \in B_{r_0}(x_0)$ and $B \in \mathcal{B}(\mathbb{R}^d)$, we have

$$p(t, x, B) \geq \int_{B \cap B_{r_0}(x_0)} q(t, x, y)dy.$$

Define $\psi(\cdot) := \lambda(\cdot \cap B_{r_0}(x_0))$, where $\lambda(dx)$ denotes the Lebesgue measure on $\mathbb{R}^d$. Clearly, $\psi(dx)$ is a σ -finite measure whose support has a non-empty interior. It remains to prove that

$$\int_0^\infty p(t, x, B)dt > 0, \quad x \in \mathbb{R}^d,$$

whenever $\psi(B) > 0$, $B \in \mathcal{B}(\mathbb{R}^d)$. For $x \in B_{r_0}(x_0)$ the implication holds trivially. Let $x \in B_{r_0}^c(x_0)$ and $B \in \mathcal{B}(\mathbb{R}^d)$, $\psi(B) > 0$, be arbitrary. For all $s > 0$ we have

$$\begin{aligned} \int_0^\infty p(t, x, B)dt &\geq \int_s^\infty p(t, x, B)dt \\ &= \int_s^\infty \int_{\mathbb{R}^d} p(t-s, x, dy)p(s, y, B)dt \\ &\geq \int_s^\infty \int_{B_{r_0}(x_0)} p(t-s, x, dy)p(s, y, B)dt \\ &= \int_{B_{r_0}(x_0)} p(s, y, B) \int_s^\infty p(t-s, x, dy)dt. \end{aligned}$$

The assertion follows from the fact that $p(s, y, B) > 0$ for $y \in B_{r_0}(x_0)$, and

$$\int_s^\infty p(t-s, x, B_{r_0}(x_0))dt = \int_0^\infty p(t, x, B_{r_0}(x_0))dt = \mathbb{E}^x \left[\int_0^\infty \mathbb{1}_{\{X(t) \in B_{r_0}(x_0)\}} dt \right] > 0,$$